

Digital Electronics (II SEM B.Tech. CSE)

Couse Code: NECE102



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Digital Electronics: Course Goals and Key Takeaways



Course Objective

To develop an understanding of digital electronics and its application which will be further useful to in the advanced courses of Electronics and Industrial Engineering.

Learning Outcomes

Upon successful completion of this course, students will:

- Gain information about the basic knowledge of digital electronics to design various types of digital circuits and systems.
- Learn how to analyse the performance and working of digital circuits.
- Develop their knowledge of digital electronics for industrial and robotic applications.

Digital Electronics: Key Topics at a Glance



Unit No.	Topics to be Covered	Lecture Hours	Learning Outcome
1.	Number System, Logic Gates, Boolean Algebra, NAND and NOR Implementation of Logic Functions, Determination of Boolean Functions from Digital Circuits.	08	Acquire an understanding of number system, basic logic gates and their applications to develop digital circuits and systems.
2.	Standard Representations of Logical Functions, SOP and POS form, Simplification of Logical Functions using K-map.	08	Acquire the knowledge to represent the logic functions in standard form and its minimization.
3.	Combinational Logic Circuit: Adder, Subtractor, Magnitude Comparator, Code Converters etc. MSI Circuits: Multiplexers, Demultiplexers, Decoders, Encoders etc.	08	Understand the design and applications of combinational logic circuits.
4.	Sequential Circuits: Latches and Flip-flops, Counters, Shift Registers.	10	Acquire the ability to design and perform the analysis of different sequential circuits.
5.	Analysis of Sequential circuits: State tables and state diagrams, Timing Circuits	04	Acquire an understanding on the analysis of sequential circuits.
6.	Introduction to logic families, semiconductor memories, ROM, PAL, PLA and Microprocessor	04	Develop knowledge about the various techniques for IC designing for digital circuits.

Textbook:	Reference Books:
1. Floyd T.L., Digital Fundamentals, 10/2, Pearson Education, 2011.	1. Digital Logic and Computer Design by M. Morris Mano, Prentice –Hall, 2015. 2. Fundamental of Digital Electronics by A. Anand Kumar, 4/e Prentice Hall, 2016.

Digital Electronics: Lecture Plan



Sl. No.	Topics	No of classes (tentative)
1.	Introduction to Number Systems and Conversion Techniques	8
2.	Continue.....	
3.	1's and 2's complement	
4.	Logic Gates (AND, OR, NOT, NAND, NOR, XOR, XNOR)	
5.	Boolean Algebra: Laws and Theorems	
6.	Boolean Function Implementation Using NAND/NOR Gates	
7.	Determination of Boolean Functions from Digital Circuits	
8.	Continue.....	
9.	SOP Standard Forms of Logical Functions	8
10.	POS Standard Forms of Logical Functions	
11.	Simplification of Logical Functions Using Boolean Algebra	
12.	Introduction to Karnaugh Maps (K-Maps)	
13.	K-Map Simplification (2-variable Functions)	
14.	K-Map Simplification (3-variable Functions)	
15.	K-Map Simplification (4-variable Functions)	
16.	Don't Care Conditions in K-Map Simplification	

Digital Electronics: Lecture Plan



17.	Introduction to Combinational Logic Circuits	8
18.	Design of Adders (Half Adder, Full Adder)	
19.	Design of Subtractors (Half Subtractor, Full Subtractor)	
20.	Magnitude Comparator Design	
21.	Code Converters (Binary to Gray, Gray to Binary, BCD to Excess-3)	
22.	Multiplexers: Design and Applications	
23.	Demultiplexers: Design and Applications	
24.	Decoders and Encoders	
25.	Introduction to Sequential Circuits	10
26.	Latches: SR Latch, D Latch	
27.	Flip-Flops: SR, JK Flip-Flops	
28.	Flip-Flops: D and T Flip-Flops	
29.	Asynchronous Counters: Design and Operation	
30.	Continue.....	
31.	Synchronous Counters: Design and Operation	
32.	Continue.....	
33.	Shift Registers: Types and Applications	
34.	Continue.....	

Digital Electronics: Weightage



Weightage of Different Components

1.	Quiz/ Assignment	10 Marks
2.	Mid Semester Examination	30 Marks
3.	Quiz/ Assignment	10 Marks
4.	End Semester Examination	50 Marks
Total Marks = 100		

Example:

Expand $A(\bar{B} + A)B$ to maxterms and minterms.

$$A = A + B\bar{B} = (A + B)(A + \bar{B})$$

$$B = B + A\bar{A} = (B + A)(B + \bar{A})$$

or

$$A(\bar{B} + A)B = (A + B)(A + \bar{B})(A + \bar{B})(A + B)(\bar{A} + B)$$

$$= (A + B)(A + \bar{B})(\bar{A} + B)$$

$$= (00)(01)(10)$$

$$= M_0 \cdot M_1 \cdot M_2$$

$$= \Pi M(0, 1, 2)$$

The maxterm M_3 is missing in the POS form. So, the SOP form will contain only the minterm m_3 .

Example:

Expand $A(\bar{A} + B)(\bar{A} + B + \bar{C})$ to maxterms and minterms.

$$\begin{aligned} A &= A + B\bar{B} + C\bar{C} = (A + B)(A + \bar{B}) + C\bar{C} \\ &= (A + B + C\bar{C})(A + \bar{B} + C\bar{C}) \\ &= (A + B + C)(A + B + \bar{C})(A + \bar{B} + C)(A + \bar{B} + \bar{C}) \\ \bar{A} + B &= \bar{A} + B + C\bar{C} = (\bar{A} + B + C)(\bar{A} + B + \bar{C}) \end{aligned}$$

Therefore,

$$\begin{aligned} A(\bar{A} + B)(\bar{A} + B + \bar{C}) &= (A + B + C)(A + B + \bar{C})(A + \bar{B} + C)(A + \bar{B} + \bar{C})(\bar{A} + B + C)(\bar{A} + B + \bar{C}) \\ &= (000)(001)(010)(011)(100)(101) \\ &= M_0 \cdot M_1 \cdot M_2 \cdot M_3 \cdot M_4 \cdot M_5 \\ &= \Pi M(0, 1, 2, 3, 4, 5) \end{aligned}$$

The maxterms M_6 and M_7 are missing in the POS form. So, the SOP form will contain the minterms 6 and 7. Therefore, the given expression in the SOP form is $\Sigma m(6, 7)$.

Boolean Algebra



Example: Write the algebraic terms of a four-variable expression having the

(a) m_0	(b) m_5	(c) m_9	(d) m_{14}	
Given minterm	m_0	m_5	m_9	m_{14}
Binary form	0000	0101	1001	1110
Product term	$\overline{A}\overline{B}\overline{C}\overline{D}$	$\overline{A}B\overline{C}D$	$A\overline{B}\overline{C}D$	$ABC\overline{D}$

Example: Write the algebraic terms of a four-variable expression having the following maxterms

(a) M_3	(b) M_9	(c) M_{11}	(d) M_{14}	
Given maxterm	M_3	M_9	M_{11}	M_{14}
Binary form	0011	1001	1011	1110
Sum term	$A + B + \bar{C} + \bar{D}$	$\bar{A} + B + C + \bar{D}$	$\bar{A} + B + \bar{C} + \bar{D}$	$\bar{A} + \bar{B} + \bar{C} + D$

Boolean Algebra



Example: Express the following functions as sum of minterms and product of maxterms:

(a) $F(A, B, C, D) = \bar{B}D + \bar{A}D + BD$

(a)
$$\begin{aligned} F &= \bar{B}D + \bar{A}D + BD = X\ 0\ X1 + 0XX1 + X1X1 \\ &= 0001, 0011, 1001, 1011, 0001, 0011, 0101, 0111, 0101, 0111, 1101, 1111 \\ &= \Sigma m(1, 3, 9, 11, 1, 3, 5, 7, 5, 7, 13, 15) \\ &= \Sigma m(1, 3, 5, 7, 9, 11, 13, 15) \\ &= \Pi M(0, 2, 4, 6, 8, 10, 12, 14) \end{aligned}$$

(b) $F(x, y, z) = (xy + z)(xz + y)$

(b)
$$\begin{aligned} F &= (xy + z)(xz + y) = (x + z)(y + z)(x + y)(y + z) \\ &= (x + y)(x + z)(y + z) \\ &= (00X)(0X0)(X00) \\ &= (000)(001)(000)(010)(000)(100) \\ &= \Pi M(0, 1, 0, 2, 0, 4) = \Pi M(0, 1, 2, 4) \\ &= \Sigma m(3, 5, 6, 7) \end{aligned}$$

Minimization of Switching Functions using Karnaugh Map



- ❖ Boolean expressions can be simplified algebraically, but being not a systematic method, we can never be sure whether the minimal expression obtained is the real minimal or not.
- ❖ The effectiveness of algebraic simplification depends on our familiarity with, and ability to apply Boolean algebraic rules, laws and theorems.
- ❖ The Karnaugh map (K-map) method, on the other hand, is a systematic method of simplifying the Boolean expressions.
- ❖ *The K-map is a chart or a graph, composed of an arrangement of adjacent cells, each representing a particular combination of variables in sum or product form.*
- ❖ Like a truth table, it is a means of showing the relationship between the logic inputs and the desired output.
- ❖ Although a K-map can be used for problems involving any number of variables, it becomes tedious for problems involving five or more variables. Usually, it is limited to six variables.
- ❖ An n variable function can have 2^n possible combinations of product terms in SOP form, or 2^n possible combinations of sum terms in POS form.
- ❖ Since the K-map is a graphical representation of Boolean expressions, a two-variable K-map will have $2^2 = 4$ cells or squares, a three variable map will have $2^3 = 8$ cells or squares, and a four variable map will have $2^4 = 16$ cells, and so on

Two Variable K-Map



- ❖ A two-variable expression can have $2^2 = 4$ possible combinations of the input variables A and B.
- ❖ Each of these combinations, $\bar{A}\bar{B}$, $\bar{A}B$, $A\bar{B}$, and AB (in the SOP form) is called a minterm.
- ❖ m_0 for $\bar{A}\bar{B}$, m_1 for $\bar{A}B$, m_2 for $A\bar{B}$, and m_3 for AB , assuming that A represents the MSB.

Consider the expression $f = \bar{A}\bar{B} + A\bar{B} + AB$. It can be expressed using minterms as

$$F = m_0 + m_2 + m_3 = \Sigma m (0, 2, 3)$$

Minterm	Inputs		Output f
	A	B	
0	0	0	1
1	0	1	0
2	1	0	1
3	1	1	1

The first column indicates the minterm designation, the second column indicates the input combinations, and the last column indicates the presence or absence of that minterm in the output expression.

Two Variable K-Map

Mapping of SOP Expressions

- ❖ A two-variable K-map has $2^2 = 4$ squares. These squares are called *cells*. Each square on the K-map represents a unique minterm.
- ❖ A 1 placed in any square indicates that the corresponding minterm is included in the output expression, and a 0 or no entry in any square indicates that the corresponding minterm does not appear in the expression for output

		B	
		0	1
A	0	$\bar{A}\bar{B}$ ⁰	$\bar{A}B$ ¹
	1	$A\bar{B}$ ²	AB ³

Minterms of a two-variable K-map

		B	
		0	1
A	0	1 ⁰	0 ¹
	1	1 ²	1 ³

The mapping of the expression $\Sigma m(0, 2, 3)$

Two Variable K-Map

Mapping of SOP Expressions

Example- Map the Expression:

$$f = \bar{A}B + A\bar{B}$$

The given expression in minterms is

$$f = m_1 + m_2 = \Sigma m(1, 2)$$

		B	
		0	1
A	0	$\bar{A}\bar{B}^0$	$\bar{A}B^1$
	1	$A\bar{B}^2$	AB^3

		B	
		0	1
A	0	0	1
	1	1	0

Two Variable K-Map

Minimization of SOP Expressions

- ❖ To minimize a Boolean expression given in the SOP form by using the K-map, we have to look for adjacent squares having 1s, that is, minterms adjacent to each other, and combine them to form larger squares to eliminate some variables.
- ❖ **Two squares are said to be adjacent to each other, if their minterms differ in only one variable.**
- ❖ For example, in a two-variable K-map, m_0 and m_1 , i.e. $\bar{A}\bar{B}$ and $\bar{A}B$ differ only in variable B (A is common to both of them). So, they may be combined to form a 2-square to eliminate the variable B.
- ❖ Similarly, minterms m_0 ($\bar{A}\bar{B}$) and m_2 ($A\bar{B}$); m_1 ($\bar{A}B$) and m_3 (AB); and m_2 ($A\bar{B}$) and m_3 (AB) are adjacent to each other.
- ❖ However, minterms m_0 ($\bar{A}\bar{B}$) and m_3 (AB), and m_1 ($\bar{A}B$) and m_2 ($A\bar{B}$) are not adjacent to each other, because they differ in more than one variable

m_0 and m_1 can be combined to yield,

$$f_1 = m_0 + m_1 = \bar{A}\bar{B} + \bar{A}B = \bar{A}(B + \bar{B}) = \bar{A}$$

m_0 and m_2 can be combined to yield,

$$f_2 = m_0 + m_2 = \bar{A}\bar{B} + A\bar{B} = \bar{B}(A + \bar{A}) = \bar{B}$$

m_1 and m_3 can be combined to yield,

$$f_3 = m_1 + m_3 = \bar{A}B + AB = B(A + \bar{A}) = B$$

m_2 and m_3 can be combined to yield,

$$f_4 = m_2 + m_3 = A\bar{B} + AB = A(B + \bar{B}) = A$$

$m_0, m_1, m_2,$ and m_3 can be combined to yield, $f_5 = m_0 + m_1 + m_2 + m_3$

$$\begin{aligned} &= \bar{A}\bar{B} + \bar{A}B + A\bar{B} + AB \\ &= \bar{A}(B + \bar{B}) + A(B + \bar{B}) \\ &= \bar{A} + A \\ &= 1 \end{aligned}$$

		B	
		0	1
A	0	1	1
	1	0	0

$f_1 = \bar{A}$

		B	
		0	1
A	0	1	0
	1	1	0

$f_2 = \bar{B}$

		B	
		0	1
A	0	0	1
	1	0	1

$f_3 = B$

		B	
		0	1
A	0	0	0
	1	1	1

$f_4 = A$

		B	
		0	1
A	0	1	1
	1	1	1

$f_5 = 1$

The possible minterm groupings in a two-variable K-map.

To read the squares on the map after minimization, consider only those variables which remain constant throughout the square, and ignore the variables which are varying.

Two Variable K-Map

Minimization of SOP Expressions

Example- Reduce the expression using Mapping:

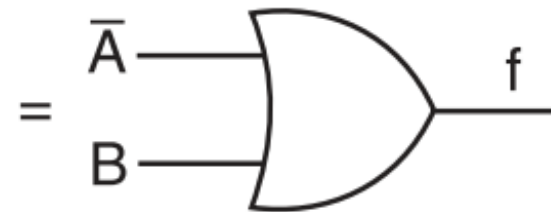
$$f = \bar{A}\bar{B} + \bar{A}B + AB$$

Expressed in terms of minterms, the given expression is

$$f = m_0 + m_1 + m_3 = \Sigma m (0, 1, 3)$$

		B	
		0	1
A	0	0 1	1 1
	1	0 2	1 3

$$f = \bar{A} + B$$



Two Variable K-Map

Mapping of POS Expressions

- ❖ Each sum term in the standard POS expression is called a *maxterm*.
- ❖ A function in two variables (A, B) has four possible maxterms, $A + B$; $A + \bar{B}$; $\bar{A} + B$; $\bar{A} + \bar{B}$
- ❖ They are represented as M_0 , M_1 , M_2 , and M_3 respectively.
- ❖ For mapping a POS expression on to the K-map, 0s are placed in the squares corresponding to the maxterms which are present in the expression and 1s are placed (or no entries are made) in the squares corresponding to the maxterms which are not present in the expression.

		B	
		0	1
A	0	⁰ $A + B$	¹ $A + \bar{B}$
	1	² $\bar{A} + B$	³ $\bar{A} + \bar{B}$

The maxterms of a two-variable K-map.

Two Variable K-Map

Mapping of POS Expressions

Example: Plot the expression on the K-map

$$f = (A + B)(\bar{A} + B)(\bar{A} + \bar{B})$$

		B	
		0	1
A	0	0 $A + B$	1 $A + \bar{B}$
	1	2 $\bar{A} + B$	3 $\bar{A} + \bar{B}$

The maxterms of a two-variable K-map.

		B	
		0	1
A	0	0	1
	1	0	0

K-Map for $\Pi M (0, 2, 3)$.

Two Variable K-Map

Minimization of POS Expressions

- ❖ To obtain the minimal expression in POS form, map the given POS expression on to the K-map and combine the adjacent 0s into as large squares as possible.
- ❖ Read the squares putting the complemented variable if its value remains constant as a 1 and the non-complemented variable if its value remains constant as a 0 along the entire square (ignoring the variables which do not remain constant throughout the square) and then write them as a sum term.

		B	
		0	1
A	0	0	1
	1	1	1

$$f_1 = A$$

		B	
		0	1
A	0	1	0
	1	1	0

$$f_2 = \bar{B}$$

		B	
		0	1
A	0	0	1
	1	0	1

$$f_3 = B$$

		B	
		0	1
A	0	1	0
	1	0	0

$$f_4 = \bar{A}$$

		B	
		0	1
A	0	0	0
	1	0	0

$$f_5 = 0$$

The possible maxterm groupings in a two-variable K-map

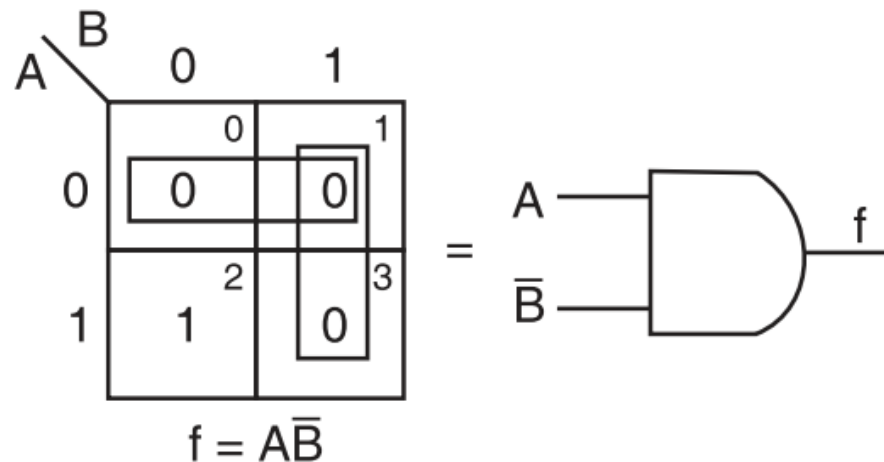
Two Variable K-Map

Minimization of POS Expressions

Example: Reduce the expression using mapping

$$f = (A + B)(A + \bar{B})(\bar{A} + \bar{B})$$

The given expression in terms of maxterms is $f = \Pi M(0, 1, 3)$.



Three Variable K-Map

- ❖ A function in three variables (A, B, C) expressed in the standard SOP form can have eight possible combinations:
 $\bar{A}\bar{B}\bar{C}, \bar{A}\bar{B}C, \bar{A}B\bar{C}, \bar{A}BC, A\bar{B}\bar{C}, A\bar{B}C, AB\bar{C}, \text{ and } ABC$
- ❖ Each one of these combinations designated by $m_0, m_1, m_2, m_3, m_4, m_5, m_6,$ and m_7 , respectively, is called a minterm.
- ❖ A is the MSB of the minterm designator and C is the LSB.
- ❖ In the standard POS form, the eight possible combinations are:
 $A + B + C, A + B + \bar{C}, A + \bar{B} + C, A + \bar{B} + \bar{C}, \bar{A} + B + C, \bar{A} + B + \bar{C}, \bar{A} + \bar{B} + C, \text{ and } \bar{A} + \bar{B} + \bar{C}$
- ❖ Each one of these combinations designated by $M_0, M_1, M_2, M_3, M_4, M_5, M_6,$ and M_7 , respectively, is called a Maxterm.
- ❖ A is the MSB of the maxterm designator and C is the LSB.

A three-variable K-map has, therefore, $8(= 2^3)$ squares or cells, and each square on the map represents a minterm or maxterm

BC		A			
		00	01	11	10
A	0	$\bar{A}\bar{B}\bar{C}$ ⁰ (m_0)	$\bar{A}\bar{B}C$ ¹ (m_1)	$\bar{A}B\bar{C}$ ³ (m_3)	$\bar{A}BC$ ² (m_2)
	1	$A\bar{B}\bar{C}$ ⁴ (m_4)	$A\bar{B}C$ ⁵ (m_5)	ABC ⁷ (m_7)	$AB\bar{C}$ ⁶ (m_6)

(a) Minterms

BC		A			
		00	01	11	10
A	0	$A + B + C$ ⁰ (M_0)	$A + B + \bar{C}$ ¹ (M_1)	$A + \bar{B} + \bar{C}$ ³ (M_3)	$A + \bar{B} + C$ ² (M_2)
	1	$\bar{A} + B + C$ ⁴ (M_4)	$\bar{A} + B + \bar{C}$ ⁵ (M_5)	$\bar{A} + \bar{B} + \bar{C}$ ⁷ (M_7)	$\bar{A} + \bar{B} + C$ ⁶ (M_6)

(b) Maxterms

The three-variable K-map.

Three Variable K-Map



❖ Example: Map the expression

$$f = \bar{A}\bar{B}C + A\bar{B}C + \bar{A}B\bar{C} + AB\bar{C} + ABC.$$

In the given expression, the minterms are: $\bar{A}\bar{B}C = 001 = m_1$; $A\bar{B}C = 101 = m_5$; $\bar{A}B\bar{C} = 010 = m_2$; $AB\bar{C} = 110 = m_6$; $ABC = 111 = m_7$. So the expression is $f = \Sigma m(1, 5, 2, 6, 7) =$

		BC			
		00	01	11	10
A	0	0 ⁰ 0	1 ¹ 1	0 ³ 0	1 ² 1
	1	0 ⁴ 0	1 ⁵ 1	1 ⁷ 1	1 ⁶ 1

K-map in SOP form.

Three Variable K-Map



❖ Example: Map the expression

$$f = (A + B + C)(\bar{A} + B + \bar{C})(\bar{A} + \bar{B} + \bar{C})(A + \bar{B} + \bar{C})(\bar{A} + \bar{B} + C)$$

In the given expression the maxterms are: $A + B + C = 000 = M_0$; $\bar{A} + B + \bar{C} = 101 = M_5$; $\bar{A} + \bar{B} + \bar{C} = 111 = M_7$; $A + \bar{B} + \bar{C} = 011 = M_3$; $\bar{A} + \bar{B} + C = 110 = M_6$.

So the expression is $f = \Pi M(0, 5, 7, 3, 6) = \Pi M(0, 3, 5, 6, 7)$. The mapping of the

BC					
		00	01	11	10
A	0	0 ⁰	1 ¹	0 ³	1 ²
	1	1 ⁴	0 ⁵	0 ⁷	0 ⁶

Three Variable K-Map



Minimization of SOP and POS Expressions

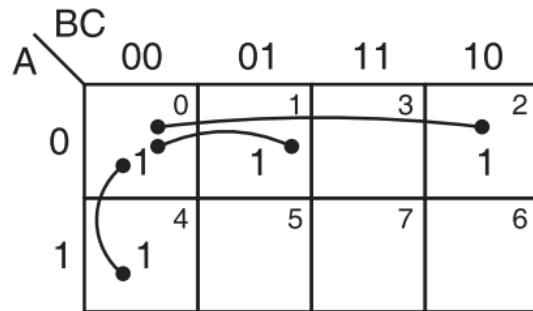
The generalized procedure to simplify the Boolean expressions as follows:

1. Plot the K-map and place 1s (0s) corresponding to the minterms (maxterms) of the SOP(POS) expression.
2. Check the K-map for 1s (0s) which are not adjacent to any other 1 (0). They are isolated minterms (maxterms). They are to be read as they are because they cannot be combined even into a 2-square.
3. Check for those 1s (0s) which are adjacent to only one other 1 (0) and make them pairs (2 squares).
4. Check for quads (4 squares) and octets (8 squares) of adjacent 1s (0s) even if they contain some 1s (0s) which have already been combined. They must geometrically form a square or a rectangle.
5. Check for any 1s (0s) that have not been combined yet and combine them into bigger squares if possible.
6. Form the minimal expression by summing (multiplying) the product (sum) terms of all the groups.

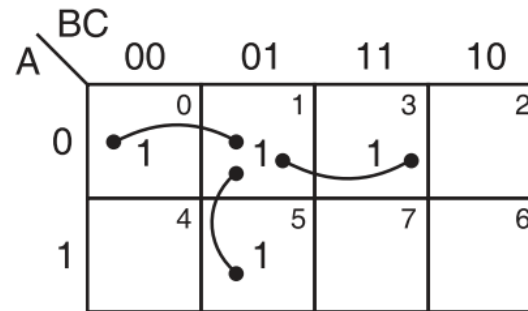
Three Variable K-Map



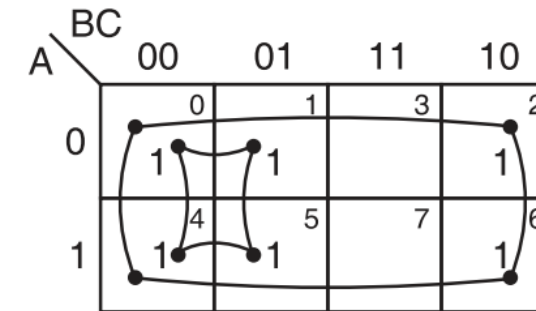
Reading the K-maps



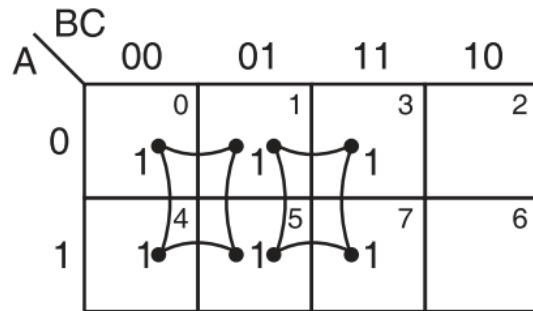
$$f_1 = \bar{B}\bar{C} + \bar{A}\bar{B} + \bar{A}\bar{C}$$



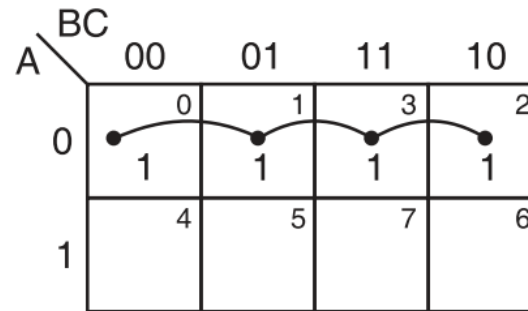
$$f_2 = \bar{A}\bar{B} + \bar{B}C + \bar{A}C$$



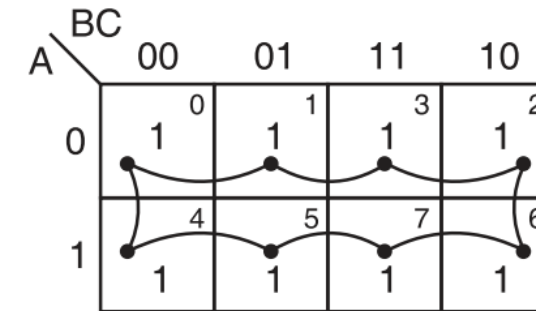
$$f_3 = \bar{C} + \bar{B}$$



$$f_4 = \bar{B} + C$$



$$f_5 = \bar{A}$$



$$f_6 = 1$$

Some possible combinations of minterms in a three-variable K-map (SOP form).

$$\begin{aligned} f_5 &= m_0 + m_1 + m_2 + m_3 = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}B\bar{C} + \bar{A}BC \\ &= \bar{A}\bar{B}(\bar{C} + C) + \bar{A}B(\bar{C} + C) \\ &= \bar{A}\bar{B} + \bar{A}B = \bar{A}(\bar{B} + B) = \bar{A} \end{aligned}$$

Three Variable K-Map

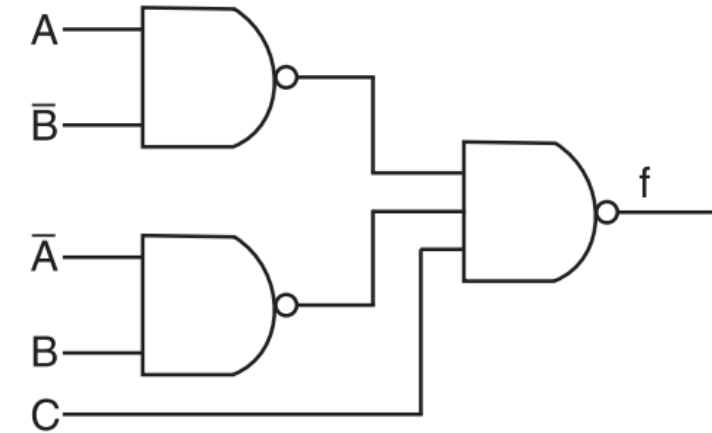
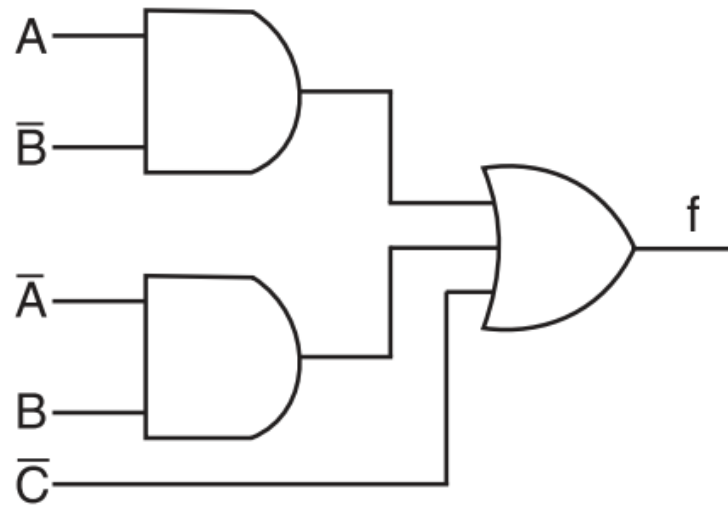


❖ Example: Reduce the expression using mapping and implement the reduced expression using NAND gates

$$f = \sum m(0, 2, 3, 4, 5, 6)$$

A \ BC	00		01		11		10	
	0	1	2	3	4	5	6	7
0	1				1		1	
1		1		1				1

$f = \bar{C} + A\bar{B} + \bar{A}B$



$$f_{\min} = A\bar{B} + \bar{A}B + \bar{C} = \overline{\overline{A\bar{B} + \bar{A}B + \bar{C}}} = \overline{\overline{A\bar{B}} \cdot \overline{\bar{A}B} \cdot \overline{\bar{C}}} = \overline{\overline{A\bar{B}} \cdot \overline{\bar{A}B} \cdot C}$$

Three Variable K-Map

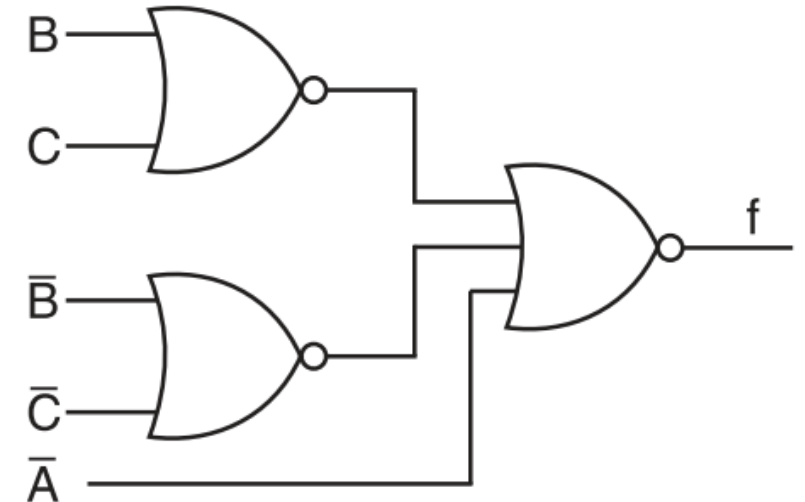
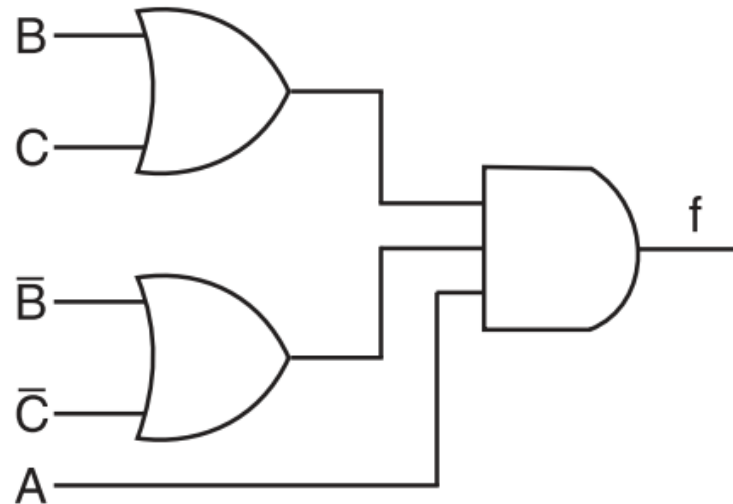


❖ Example: Reduce the expression using mapping and implement the reduced expression using NOR gates

$$f = \Pi M (0, 1, 2, 3, 4, 7)$$

A \ BC	00		01		11		10	
	0	1	2	3	4	5	6	7
0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0

$f = (B + C)(\bar{B} + \bar{C})(A)$



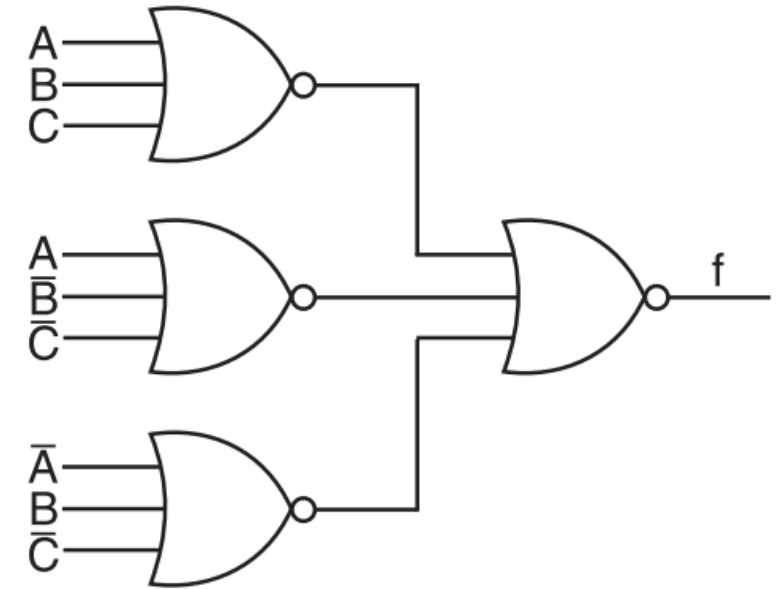
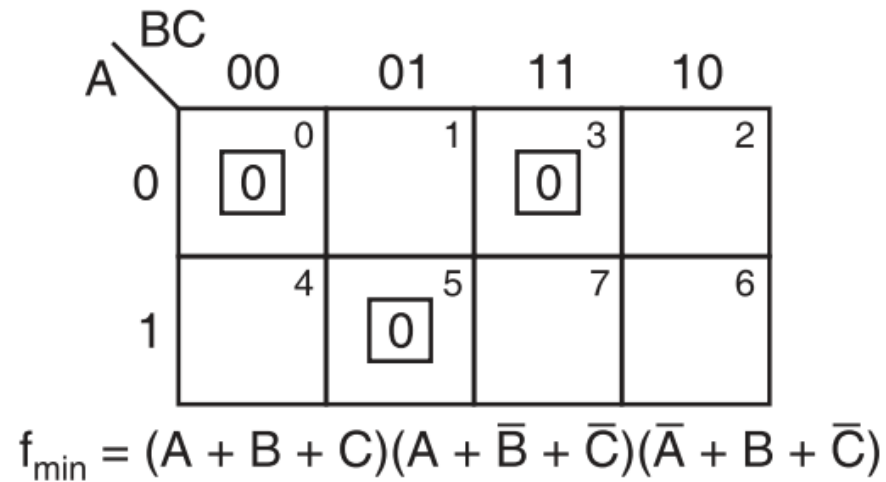
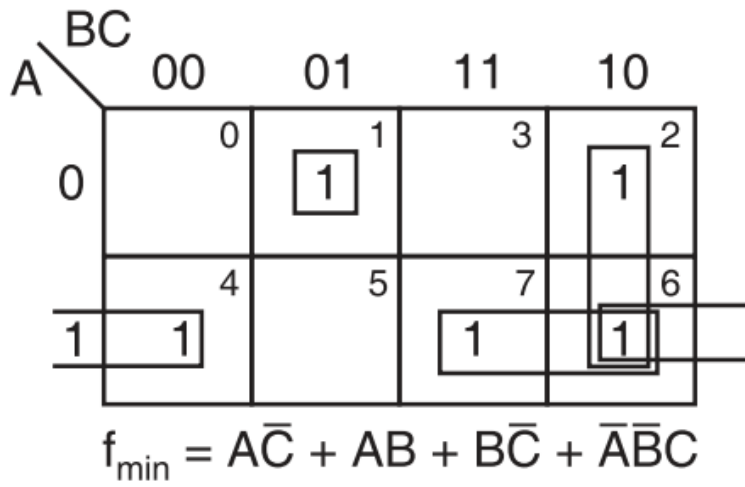
$$f_{\min} = (B + C)(\bar{B} + \bar{C})(A) = \overline{\overline{(B + C)(\bar{B} + \bar{C})(A)}} = \overline{\overline{(B + C)} + \overline{\overline{(\bar{B} + \bar{C})}} + \overline{(\bar{A})}}$$

Three Variable K-Map



Example: Obtain the real minimal expression and implement it using universal gates.

$$f = \sum m(1, 2, 4, 6, 7)$$



Three Variable K-Map



Example: Show the truth table for each of the following functions and find its simplest POS form

(a) $f(X, Y, Z) = XY + XZ$

X	Y	Z	XY	XZ	XY + XZ
0	0	0	0	0	$0 + 0 = 0$
0	0	1	0	0	$0 + 0 = 0$
0	1	0	0	0	$0 + 0 = 0$
0	1	1	0	0	$0 + 0 = 0$
1	0	0	0	0	$0 + 0 = 0$
1	0	1	0	1	$0 + 1 = 1$
1	1	0	1	0	$1 + 0 = 1$
1	1	1	1	1	$1 + 1 = 1$

POS form $f = \Pi M(0, 1, 2, 3, 4)$

X \ YZ				
	00	01	11	10
0	0	0	0	0
1	0			

$$f_{\min} = X(Y + Z)$$

Three Variable K-Map



Example: Show the truth table for each of the following functions and find its simplest POS form

(b) $f(X, Y, Z) = \bar{X} + Y\bar{Z}$

X	Y	Z	$\bar{X}$	$\bar{Z}$	$Y\bar{Z}$	$\bar{X} + Y\bar{Z}$
0	0	0	1	1	0	$1 + 0 = 1$
0	0	1	1	0	0	$1 + 0 = 1$
0	1	0	1	1	1	$1 + 1 = 1$
0	1	1	1	0	0	$1 + 0 = 1$
1	0	0	0	1	0	$0 + 0 = 0$
1	0	1	0	0	0	$0 + 0 = 0$
1	1	0	0	1	1	$0 + 1 = 1$
1	1	1	0	0	0	$0 + 0 = 0$

in POS form $f = \Pi M(4, 5, 7)$

X \ YZ				
	00	01	11	10
0	0	1	3	2
1	4 0	5 0	7 0	6

$$f_{\min} = (\bar{X} + Y)(\bar{X} + \bar{Z})$$

Four Variable K-Map

A four-variable (A, B, C, D) expression can have $2^4 = 16$ possible combinations of input variables such as

$\bar{A}\bar{B}\bar{C}\bar{D}$, $\bar{A}\bar{B}\bar{C}D$, ..., $ABCD$ with minterm designations $m_0, m_1, \dots, m_{15}$, respectively,

$A + B + C + D$, $A + B + C + \bar{D}$, ..., $\bar{A} + \bar{B} + \bar{C} + \bar{D}$ with maxterm designations $M_0, M_1, \dots, M_{15}$, respectively, in the POS form.

A four-variable K-map has $2^4 = 16$ squares or cells and each square on the map represents either a minterm or a maxterm

AB \ CD	00	01	11	10
00	$\bar{A}\bar{B}\bar{C}\bar{D}$ (m_0)	$\bar{A}\bar{B}\bar{C}D$ (m_1)	$\bar{A}\bar{B}C\bar{D}$ (m_3)	$\bar{A}\bar{B}CD$ (m_2)
01	$\bar{A}B\bar{C}\bar{D}$ (m_4)	$\bar{A}B\bar{C}D$ (m_5)	$\bar{A}BC\bar{D}$ (m_7)	$\bar{A}BCD$ (m_6)
11	$AB\bar{C}\bar{D}$ (m_{12})	$AB\bar{C}D$ (m_{13})	$ABCD$ (m_{15})	$ABC\bar{D}$ (m_{14})
10	$A\bar{B}\bar{C}\bar{D}$ (m_8)	$A\bar{B}\bar{C}D$ (m_9)	$A\bar{B}C\bar{D}$ (m_{11})	$A\bar{B}CD$ (m_{10})

SOP form

AB \ CD	00	01	11	10
00	$A + B + C + D$ (M_0)	$A + B + C + \bar{D}$ (M_1)	$A + B + \bar{C} + \bar{D}$ (M_3)	$A + B + \bar{C} + D$ (M_2)
01	$A + \bar{B} + C + D$ (M_4)	$A + \bar{B} + C + \bar{D}$ (M_5)	$A + \bar{B} + \bar{C} + \bar{D}$ (M_7)	$A + \bar{B} + \bar{C} + D$ (M_6)
11	$\bar{A} + \bar{B} + C + D$ (M_{12})	$\bar{A} + \bar{B} + C + \bar{D}$ (M_{13})	$\bar{A} + \bar{B} + \bar{C} + \bar{D}$ (M_{15})	$\bar{A} + \bar{B} + \bar{C} + D$ (M_{14})
10	$\bar{A} + B + C + D$ (M_8)	$\bar{A} + B + C + \bar{D}$ (M_9)	$\bar{A} + B + \bar{C} + \bar{D}$ (M_{11})	$\bar{A} + B + \bar{C} + D$ (M_{10})

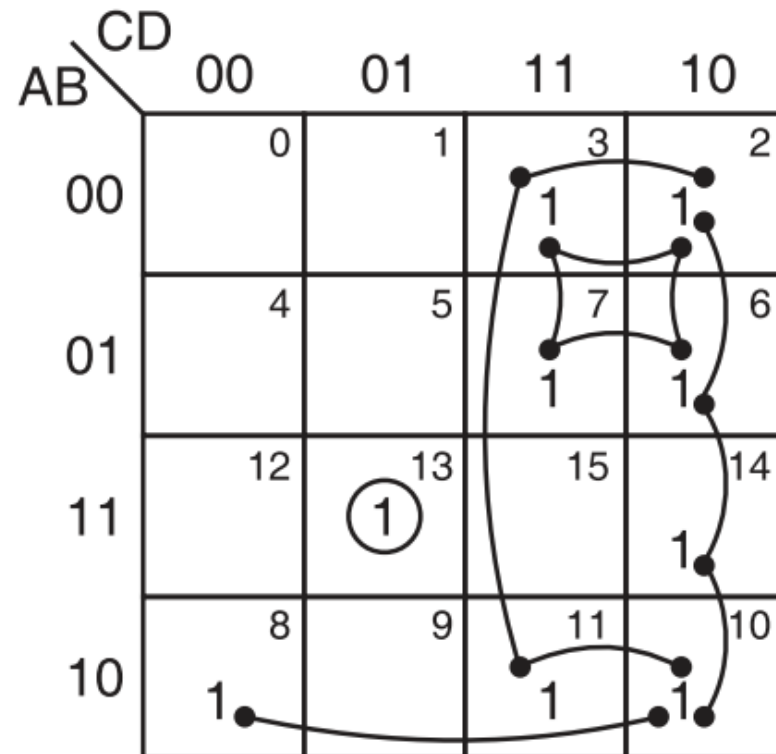
POS form

Four Variable K-Map



Example: Reduce using mapping the expression

$$f = \sum m(2, 3, 6, 7, 8, 10, 11, 13, 14).$$



$$f = ABCD + ABD + AC + BC + CD$$

Four Variable K-Map



Example: Reduce using mapping the expression:

$$f = \sum m(0, 1, 3, 5, 7, 8, 9, 10, 12, 13)$$

CD \ AB	00	01	11	10
00	0 1	1 1	3 1	2 1
01	4	5 1	7 1	6
11	12 1	13 1	15	14
10	8 1	9 1	11	10 1

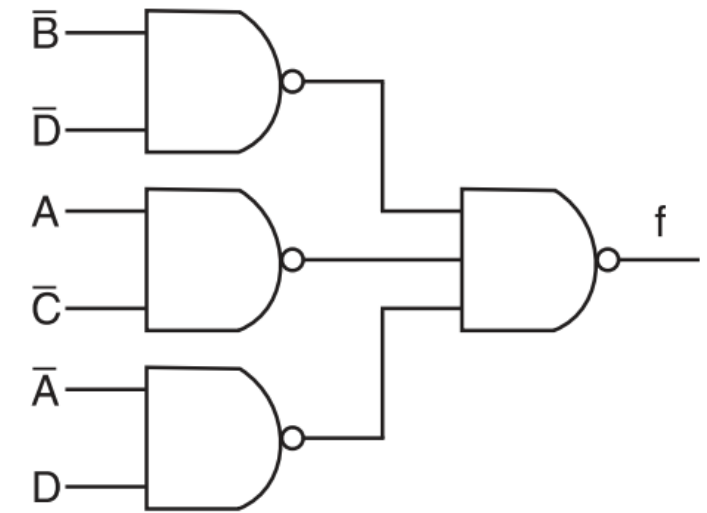
$$f_{\min} = \bar{B}\bar{D} + A\bar{C} + \bar{A}D$$

(a) SOP K-map

CD \ AB	00	01	11	10
00	0	1	3	2
01	4 0	5	7	6 0
11	12	13	15 0	14 0
10	8	9	11 0	10

$$f_{\min} = (A + \bar{B} + D)(\bar{A} + \bar{C} + \bar{D})(\bar{A} + \bar{B} + \bar{C})$$

(b) POS K-map



(c) NAND logic

Four Variable K-Map



Example: Reduce using mapping the expression and implement the real minimal expression in universal logic

$$f = \prod M(2, 8, 9, 10, 11, 12, 14)$$

CD \ AB	00	01	11	10
00	0 1	1 1	3 1	2
01	4 1	5 1	7 1	6 1
11	12	13 1	15 1	14
10	8	9	11	10

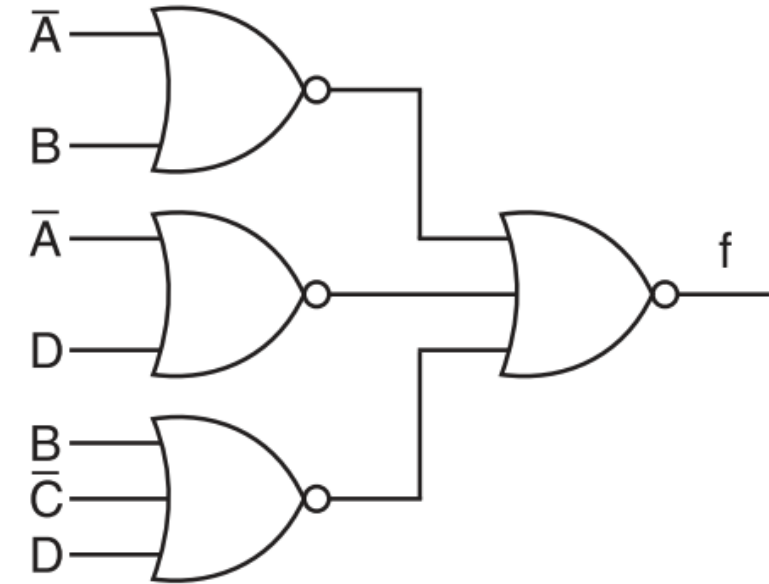
$$f_{\min} = \bar{A}\bar{C} + \bar{A}D + \bar{A}B + BD$$

(a) SOP K-map

CD \ AB	00	01	11	10
00	0	1	3	2 0
01	4	5	7	6
11	12 0	13	15	14 0
10	8 0	9 0	11 0	10 0

$$f_{\min} = (\bar{A} + B)(\bar{A} + D)(B + \bar{C} + D)$$

(b) POS K-map



(c) NOR logic

Four Variable K-Map



Example: Reduce using mapping the expression and implement the real minimal expression in universal logic

$$f = \sum m(0, 2, 4, 6, 7, 8, 10, 12, 13, 15)$$

$$f_{\min} = (B + \bar{D})(A + C + \bar{D})(\bar{A} + \bar{B} + \bar{C} + D) = \overline{(B + \bar{D}) + (A + C + \bar{D}) + (\bar{A} + \bar{B} + \bar{C} + D)}$$

CD \ AB	00	01	11	10
00	0 1	1	3	2 1
01	4 1	5	7 1	6 1
11	12 1	13 1	15 1	14
10	8 1	9	11	10 1

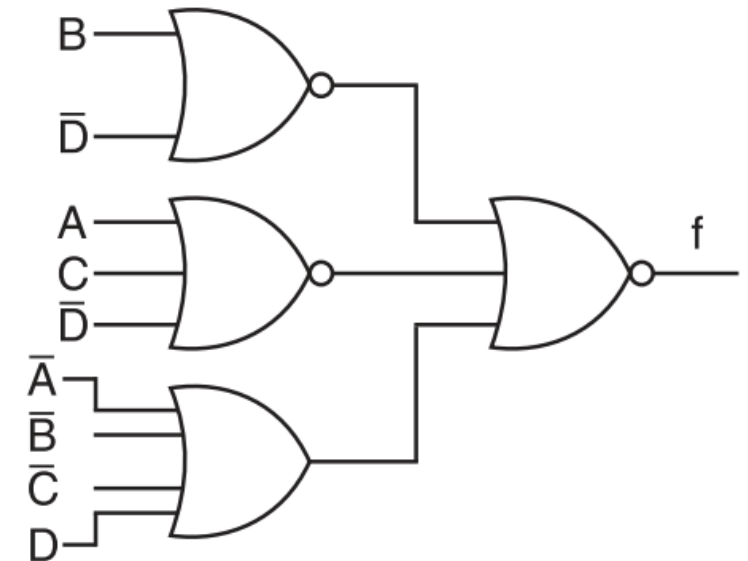
$$f_{\min} = \bar{C}\bar{D} + \bar{A}\bar{D} + \bar{B}\bar{D} + ABD + BCD$$

(a) SOP K-map

CD \ AB	00	01	11	10
00	0	1 0	3 0	2
01	4	5 0	7	6
11	12	13	15	14 0
10	8	9 0	11 0	10

$$f_{\min} = (B + \bar{D})(A + C + \bar{D})(\bar{A} + \bar{B} + \bar{C} + D)$$

(b) POS K-map



(c) Logic diagram

K-Map



Example: Reduce the function $y(A,B,C) = \Sigma (0,1,3,6,7)$

Example: Reduce the function $y = \Sigma (4,5,7)$

Example: Simplify the sop function $F(A,B,C) = \Sigma (1,2,3,7)$

Example: Simply the sop function $F(A,B,C) = \Sigma (1,3,5,7)$

Example: Simply the sop function $F(A,B,C,D) = \Sigma (1,2,3,5,8,10,11,12)$

Example: Simply the sop function $F(w,x,y,z) = \Sigma (4,5,6,7,12,13,14,15)$